

Dongwook Kwon

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EDUCATION

Kwangwoon University

Master of Science in Computer Engineering, GPA 4.33 / 4.5 (98.1%)

Seoul, Korea

Mar. 2024 – Feb. 2026

University of Toronto

Graduate Research Program, Mechanical & Industrial Engineering, GPA 3.68 / 4.0

Toronto, ON, Canada

Jan. 2025 – Jul. 2025

Kwangwoon University

B.S. Electronics & Communications Engineering, GPA 3.41 / 4.5 (88.1%)

Seoul, Korea

Mar. 2019 – Feb. 2024

EXPERIENCE

AI Engineer, Intelligent Agent Technology Team

Suresoft Technologies

Dec. 2025 – Present

Seongnam, Korea

- Developing an AI validation system utilizing AI agent-based architectures for enterprise model verification.
- Designing autonomous and collaborative AI agents that verify and evaluate machine-learning models at scale.
- Building agentic AI systems to enable systematic, scalable, and reliable AI verification processes.

Research Assistant

Bio-Computing and Machine Learning Laboratory, Kwangwoon University

Aug. 2021 – Dec. 2025

Seoul, Korea

- Researched deep-learning algorithms for anomaly detection in time-series data (ECG, cyber-physical systems).
- Co-developed an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.
- Published in EMNLP 2025 (Oral) and ACL 2026; co-author on two granted Republic of Korea patents.

Visiting Researcher

LG Electronics Canada (with University of Toronto)

Jan. 2025 – Jul. 2025

Toronto, ON, Canada

- Built an LLM-driven assessment framework for evaluating performance of agentic AI systems.
- Work fed directly into peer-reviewed publications at EMNLP 2025 and ACL 2026.

AI Development Project Participant, Summer Program

Qualcomm Institute, University of California San Diego

Jul. 2022 – Aug. 2022

San Diego, CA, USA

- Built a personality-based drug-addiction prediction system using machine-learning models.
- Recognized with the program's Achievement Award.

PROJECTS

Agentic AI Evaluation Framework | *Python, LLMs, multi-agent systems*

2025 – Present

- Developed an evaluation framework for agentic AI systems leveraging LLMs to assess performance and reliability.

AI Intrusion Detection for APR1400 Nuclear Reactors | *Python, PyTorch, anomaly detection*

2022 – Present

- Built an AI-based defense system protecting nuclear power plant control networks from cyber attacks.

Quadruped Robot with Vision SLAM | *ROS, depth camera, SLAM*

Apr 2022 – Nov 2022

- Developed autonomous navigation algorithm with SLAM and depth-camera perception for a quadruped platform.

Smart Logistics with Computer Vision | *PyTorch, OpenCV, container tracking*

Apr 2021 – Nov 2021

- Logistics detection system with location tracking and stack-layer identification — Ministerial Gold Award.

PUBLICATIONS & PATENTS

- **D. Kwon[†]** et al. "CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades." *ACL 2026 — Industry Track (Poster)*. [†]Equal contribution.

- **D. Kwon**[†] et al. “GEMMAS: Graph-based Evaluation Metrics for Multi-Agent Systems.” *EMNLP 2025* — *Industry Track (Oral)*. [†]Equal contribution.
- **D. Kwon** et al. “Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments.” Under review at *DMKD* (Springer Nature), 2025.
- **D. Kwon** et al. “Hybrid GAT + Transformer for implantable-device anomaly detection.” *IBEC 2024*. **Best Poster Silver**.
- Patent **KR 10-2961984** — Real-time anomaly detection for implantable medical devices (Apr 2026).
- Patent **KR 10-2848814** — GPT API-Based Network Anomaly Detection (Aug 2025).

TECHNICAL SKILLS

Languages: Python, C/C++, MATLAB, Bash

Frameworks: PyTorch, TensorFlow, Transformers, LangChain, NumPy, Pandas, scikit-learn

Developer Tools: Linux, Docker, Git, CUDA, VS Code

Domains: Agentic AI, multi-agent LLM systems, GNNs, time-series anomaly detection, biomedical signals